Arithmetic formulas for Gaussian feature paths with non-Gaussian controls

ChatGPT-6 Astra Kaarel Hänni Dmitry Vaintrob

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Abstract

We study iterated tanh feature maps whose Gaussian row law may be changed on a fixed-dimensional subspace at each layer. The changed coordinates may have any finite relative entropy, need not be Gaussian, and need not have moments above two. Their subspaces may change arbitrarily between layers. The same theorem permits arbitrary Gaussian controls and quadratic tanh tilts with bounded entrywise coefficient norm. A further extension allows quadratic tilts in normalized ridge differences, uniformly when their raw coefficients diverge as anchors approach one another. If q_j is the largest incoming squared feature norm and D_j the row entropy, a globally margined Boolean readout has arithmetic-formula complexity polynomial in its elementary resources and exponential only in $\sum_j (q_j + \sqrt{D_j} + D_j)$; normalized ridge tilts require only a uniform small bound on their represented coefficient mass. Logarithmic total budget therefore gives nonuniform PNC^1, hence nonuniform logarithmic space. The proof uses exact scalar Poisson formulas, Gaussian Hermite features, and a separately maintained squared-norm oracle. It includes finite-bit control for correlated Taylor subtractions and continuous feature coordinates. Positive Poisson mixtures supply the quadratic-tilt extensions without a dimension bound. Arbitrary finite-entropy laws are also allowed at the first and last layers, with an explicit endpoint cost that is constant at bounded action and fixed margin. This is a restricted feature-path theorem: it does not classify the arbitrary conditional row laws of the full critical Bayesian posterior.

1 The result and its scope

A Gaussian feature layer sends a vector u to the random-variable feature $\tanh(G(u))$, where G is an isonormal Gaussian on its feature Hilbert space. Its inner product with another query feature is the corresponding Gaussian tanh kernel. We permit a non-Gaussian change on a subspace of dimension at most a fixed integer r_0.

Precisely, at layer j choose an isometry $A_j : \mathbb{R}^{r_j} \to \mathcal{H}_{j-1}$, with $r_j \le r_0$, a law $\rho_j \ll \gamma_{r_j}$, and $Z_j \sim \rho_j$. Independently let G_j be isonormal on $\mathrm{ran}(A_j)^{\perp}$. Set

$$u_j(x)(Z_j, G_j) = \tanh\big(Z_j^* A_j^* u_{j-1}(x) + G_j((I - A_j A_j^*) u_{j-1}(x))\big), \tag{1}$$

as a vector in the appropriate L^2 space. The subspaces and laws are fixed across queries, but can change with the layer. Initialize

$$d \ge 1, \qquad X = \{-1, 1\}^d, \qquad u_0(x) = (1, x)/\sqrt{d}.$$

Write

$$D_j = \mathrm{KL}(\rho_j \| \gamma_{r_j}) < \infty, \qquad q_j = \max_{x \in X} \|u_{j-1}(x)\|^2, \qquad B = \sum_{j=1}^{H} (q_j + \sqrt{D_j} + D_j).$$

Thus $q_1 \le 2$ and subsequent physical squared norms are at most one.

Theorem 1 (Fixed-rank entropy controls)**.** *Fix r_0. Suppose $v \in \mathcal{H}_H$ and $\eta > 0$ satisfy*

$$f(x)\langle v, u_H(x)\rangle \geq \eta \qquad (x \in X)$$

for a Boolean function f. Then f is a majority of signs of rational arithmetic formulas of total size at most

$$\mathrm{poly}_{r_0}(d + H + 1 + \|v\|/\eta)\exp(C_{r_0}B). \tag{2}$$

The formulas have coefficients of polynomial bit length in this size. Consequently, when the resources in (2) are polynomial in an ambient parameter, f belongs to nonuniform PNC^1*, and hence to L/poly.*

The theorem also holds if arbitrary layers are replaced by either of the following, with their actual Gaussian relative entropies included in B:

- an arbitrary nonsingular Gaussian row law, with no rank restriction on its mean or covariance;
- a law proportional to

$$\exp\left(\frac{1}{2}\sum_{i,j} T_{ij}\tanh(G(w_i))\tanh(G(w_j))\right)d\gamma, \qquad \sum_{i,j}|T_{ij}| \leq b_0,$$

  where T is symmetric, b_0 is fixed, and $\|w_i\|^2 \leq q_j$. There is no bound on the number or span of the w_i.

For the second extension the constant may depend on b_0. Its local cost is in fact $1 + O_{b_0}(q_j)$, and its entropy is $O_{b_0}(q_j^2)$. Sections 9–10 prove these extensions using the same state interface.

There is also a version insensitive to small distances between anchors. For nonzero denominators, allow normalized features

$$\psi_i(g) = \frac{\tanh(g^*w_i)}{\|w_i\|} \quad \text{or} \quad \psi_i(g) = \frac{\tanh(g^*w_i) - \tanh(g^*z_i)}{\|w_i - z_i\|}, \qquad \|w_i\|^2, \|z_i\|^2 \leq q_j.$$

Theorem 2 (Geometrically normalized quadratic controls)**.** *There is a universal $t_0 > 0$ such that layers with row law*

$$d\rho = Z^{-1}\exp\left(\tfrac{1}{2}\sum_{i,k} S_{ik}\psi_i\psi_k\right)d\gamma, \qquad S = S^*, \qquad t = \sum_{i,k}|S_{ik}| \leq t_0,$$

can be interspersed with the preceding layers. Their contribution to the accumulated budget in (2) is $q_j + \sqrt{D_j} + D_j$, with D_j their actual entropy. The uniform condition $t_j \leq t_0$ is not summed over layers. Local work and moment growth are $1 + O(q_j + \sqrt{D_j} + D_j)$ and $D_j = O(t_j^2)$. Constants are independent of the number, span, and mutual distances of the anchors.

Section 11 proves this claim. It is a representation condition on the quadratic potential, not an assertion that every covariance-normalized quadratic form has such a representation with small t.

The conclusion is nonuniform and existential. It does not give an efficient algorithm for sampling or evaluating the densities ρ_j. It imposes no higher-moment, smoothness, lower coefficient-size, or density-description assumption. The constant may depend on r_0; the theorem is not dimension-free in the rank of the changed block.

An arbitrary finite-entropy law on the whole row space need not be a fixed-rank independent block change. In particular, the entropy chain rule produces *conditional* scalar laws, generally depending on all perpendicular coordinates. Such changes at internal layers are outside (1). The first and last layers have additional structure that permits the following extension.

Corollary 3 (Arbitrary endpoint row laws)**.** *Let $H \geq 2$. Permit an arbitrary first row law of entropy D_1 relative to γ_{d+1}, and an arbitrary final row law of entropy D_H relative to its Gaussian reference. Let the internal layers belong to any of the classes above, and let B_{mid} be the sum of their respective proved budgets. If a readout of norm V has global margin $\eta > 0$, the conclusion of Theorem 1 holds with formula size at most*

$$\mathrm{poly}(d + H + D_1 + 1 + V/\eta) \exp\left[C\left(B_{\mathrm{mid}} + D_1 + \sqrt{D_H}\frac{V}{\eta} + D_H\frac{V^2}{\eta^2}\right)\right].$$

Constants may depend on the fixed block-rank and raw-coefficient bounds. For $H = 1$, the arbitrary first-layer construction alone applies, with size $\mathrm{poly}(d + D_1 + 1 + V/\eta)\exp(CD_1)$.

In particular, bounded total entropy and readout energy, fixed margin, and a logarithmic internal budget give polynomial-size formulas. For growing action, the displayed endpoint overhead must be included; it is not automatically polynomial. Section 12 proves the corollary without a density-description assumption or an optimality assumption. It does not resolve the arbitrary *internal* row laws of the original critical posterior.

We prove the one-dimensional case first. Section 8 gives the fixed-rank extension. All claims here have conventional independent AI-agent audits; this note is not Lean verified.

Reading the construction. The base theorem changes an independent block of bounded dimension; its entropy D_j is measured against the corresponding standard Gaussian. The quadratic-tilt extensions use mixtures to obtain the same scalar-formula interface, and the endpoint argument uses a different readout-density transfer. None of these operations asserts a law for finite-width posterior weights. We prove local second-moment bounds, compose them using the budget B, and only then fix and round a finite collection of formulas. The class PNC^1 denotes nonuniform signs of polynomial-size, logarithmic-depth integer arithmetic circuits.

2 Scalar formulas and entropy estimates

A random scalar formula has shape and constants chosen independently of the input. A fresh call is an independent realization. When a cancellation uses the same sample twice, its realized expression is copied syntactically and its entire size is charged twice. There is no shared vector computation.

Intermediate formulas may use complex arithmetic and real or imaginary projection. We count their size in this fixed-dimensional algebra and convert the *entire selected expression* to real arithmetic only once. Expanding every local complex product separately would give an incorrect cost for some of the estimates below.

Lemma 4 (Poisson transforms)**.** *Suppose X has mean a, second moment at most V, and expected formula size $T \geq 1$. For $K \sim \mathrm{Pois}(\lambda)$ and independent copies X_i, put*

$$Z_s = \prod_{i=1}^{K}(1 + isX_i/\lambda), \qquad \lambda = 1 + s^2V.$$

Then $\mathbb{E}Z_s = e^{isa}$, $\mathbb{E}|Z_s|^2 \leq e$, and the expected complex-formula size is $O(\lambda T)$. At expected size $O(T)$, unbiased factories for $\sin(sa)$, $\cos(sa)$, and $\cos(sa) - 1$ have second moments bounded respectively by

$$Cs^2V, \qquad C(1 + s^2V), \qquad C\min(s^4V^2, s^2V). \tag{3}$$

If $c \geq 0$ and R has mean $r \in [0, 1]$ and second moment at most a fixed constant, the real Poisson product

$$E_c = \prod_{i=1}^{K}(1 - cR_i/\lambda), \qquad \lambda = 1 + c^2V,$$

has mean e^{-cr}. At expected size $O(T)$ it has second moment at most $C(1+c^2V)e^{-2cr}$, and a factory for $e^{-cr}-1$ has second moment at most Cc^2V.

Proof. The compound-Poisson identity gives

$$\mathbb{E}Z_s = e^{isa}, \quad \mathbb{E}|Z_s|^2 = e^{s^2\mathbb{E}X^2/\lambda}, \quad \mathbb{E}E_c^2 = e^{-2cr+c^2\mathbb{E}R^2/\lambda}.$$

An estimator costing λT can be returned with probability $1/\lambda$ and multiplied by λ, giving cost $O(T)$ and multiplying its second moment by λ. Write $u = sa$ and $w = s^2\mathbb{E}X^2/\lambda \le 1$. Direct calculation gives

$$\mathbb{E}(\Re Z_s - 1)^2 = (\cos u - 1)^2 + \cos^2 u(\cosh w - 1) + \sin^2 u \sinh w.$$

Before the cost adjustment this is $O(\min(s^4V^2, 1))$. Similarly $\mathbb{E}(\Im Z_s)^2 = O(\min(s^2V, 1))$. These prove (3). Finally

$$\mathbb{E}(E_c - 1)^2 = e^{-2cr}(e^{c^2\mathbb{E}R^2/\lambda} - 1) + (e^{-cr} - 1)^2.$$

For small c this is $O(c^2V)$, and for large c it is bounded; the same cost adjustment proves the last assertion. Averaging a cheaper estimator, when needed, also preserves these work–moment bounds. □

Put $k(\omega) = 1/\sinh(\pi\omega/2)$. We use

$$\tanh t = \int_0^\infty k(\omega)\sin(\omega t)\,d\omega, \quad \int_0^\infty \omega k(\omega)\,d\omega = 1, \quad \int_0^\infty \omega^3 k(\omega)\,d\omega = 2. \tag{4}$$

These identities follow by the Fourier transform of sech^2 and differentiation or integration. The densities $\omega k(\omega)$ and $\omega^3 k(\omega)/2$ are used below; both have all polynomial moments.

Lemma 5 (Capped scalar residual)**.** *For $0 < q \le 2$, a scalar oracle X of bounded second moment and mean $a \in [-1, 1]$, and fixed $z \in \mathbb{R}$, the function*

$$\tanh(\sqrt{q}za)/\sqrt{q} - za$$

has an unbiased complex-formula factory of expected size $O(T)$ and second moment

$$C\min(q^2z^6, z^2). \tag{5}$$

If $\mathrm{KL}(\rho\|\gamma_1) = D$, then

$$\mathbb{E}_\rho \min(q^2Z^6, Z^2) \le C(q^2 + D), \tag{6}$$
$$\mathbb{E}_\rho \min(qZ^4, Z^2) \le C(q + D). \tag{7}$$

Proof. For qz^2 small, use the density $\omega^3 k(\omega)/2$ and subtract the linear term from the imaginary Poisson product using the *same* children. The resulting polynomial starts at degree three. Explicitly, with $u = sa$, $w = s^2\mathbb{E}X^2/\lambda$, and $L = (s/\lambda)\sum_i X_i$,

$$\mathbb{E}(\Im Z_s - L)^2 = (\sin u - u)^2 + w(\cos u - 1)^2 + \sin^2 u(\cosh w - 1) + \cos^2 u(\sinh w - w).$$

This is $O(s^6V^3)$ when $\lambda = 1 + s^2V$; for large s the weaker quadratic bound is already at most that sixth-order bound. After the Fourier scaling its second moment is $O(q^2z^6)$, and its expected work is $O(T)$ after integrating the frequency. For qz^2 large, the ordinary Fourier–Poisson estimator for $\phi_s(a) = \tanh(sa)/s$ has work times second moment bounded by CVT, uniformly in s. Indeed optimal frequency sampling gives the product proportional to

$$s^{-2}\left[\int_0^\infty k(\omega)\min(1, \omega|s|\sqrt{V})\sqrt{1 + \omega^2 s^2 V}\,d\omega\right]^2 \le CV.$$

Balancing work by sampling or averaging, multiplying by z, and subtracting za gives the second cap. The exact cubic-subtraction identity uses only second moments of distinct children.

For either entropy claim let $W(Z)$ denote its capped integrand. It is bounded by Z^2, and also by q^2Z^6 or qZ^4. At a fixed sufficiently small $c > 0$,

$$e^{cW} - 1 \le cWe^{cZ^2}.$$

Gaussian moments therefore give $\log \mathbb{E}_\gamma e^{cW} \le Cq^2$ or Cq. Entropy duality proves (6)–(7). □

3 One changed direction and Gaussian tensor features

For now write the normalized input as $h = \alpha e_0 + v$, with $\alpha^2 + \|v\|^2 \le 1$, and let $r = \|v\|^2$. The local feature is

$$F(h)(z, G) = \tanh(\sqrt{q}[z\alpha + G(v)])/\sqrt{q}. \tag{8}$$

Assume scalar oracles of common expected size T for α, r, and every coordinate of v, with bounded scalar second moments and a bounded *trace* of the coordinate moments. No sum of coordinatewise suprema over inputs is required.

Expand only the perpendicular Gaussian variables in their Hermite/Fock representation. The full ordered tensor coordinates of degree ℓ are indexed by $I = (i_1, \dots, i_\ell)$. Their coefficient is

$$\frac{1}{\sqrt{q}} \int_0^\infty k(\omega) e^{-q\omega^2 r/2} \frac{(\omega\sqrt{q})^\ell}{\sqrt{\ell!}} \begin{cases} \pm \sin(\omega\sqrt{q}z\alpha), & \ell \text{ even}, \\ \pm \cos(\omega\sqrt{q}z\alpha), & \ell \text{ odd}, \end{cases} v_{i_1} \cdots v_{i_\ell}\, d\omega. \tag{9}$$

The ordered tensors contain redundant symmetric coordinates but preserve the norm and kernel. Continuous coordinate measures are allowed.

Lemma 6 (Nonlinear residual)**.** *Let $L(h) = z\alpha + G(v)$ and $R = F - L$. There are coordinate oracles of expected size $O(T)$ in degrees zero and one, and expected size $O((\ell+1)T)$ in degree $\ell \ge 2$, such that*

$$\mathbb{E}_\rho \left[\mathbb{E}|R_0|^2 + \int \mathbb{E}|R_{1,i}|^2, di + \sum_{\ell \ge 2} \ell^2 \int \mathbb{E}|R_{\ell,I}|^2, dI \right] \le C(q^2 + D). \tag{10}$$

The constants are uniform for $0 < q \le 2$ and finite D.

Proof. For $\ell \ge 2$, sample ω with density $\omega^3 k(\omega)/2$. Use the relative exponential factory of Lemma 4 for the damping in (9). Average $m = \lceil C(1 + q\omega^2 V) \rceil$ independent copies of each tensor child. Its trace moment is at most $r + V/m$. The exponential damping cancels the true-r part of the tensor sum:

$$e^{-q\omega^2 r} \sum_{\ell \ge 0} \frac{[q\omega^2(r + V/m)]^\ell}{\ell!} = e^{q\omega^2 V/m} \le C.$$

For odd degrees the tail begins at degree three. For even degrees it begins at degree two, but the sine factory contributes the additional factor $Cq\omega^2 z^2$. With ℓ^2 inserted, the summed moment is thus at most $Cq^2(1 + z^2)$ after integration over the frequency. All remaining frequency factors are fixed polynomials, and the exponential tail of the proposal integrates them. The expected number of tensor children is $O(\ell)$.

The degree-one correction is

$$\int_0^\infty \omega k(\omega) \left[(\cos(\omega\sqrt{q}z\alpha) - 1)e^{-q\omega^2 r/2} + (e^{-q\omega^2 r/2} - 1) \right] d\omega.$$

Lemma 4 bounds its moment by $C[\min(q^2 z^4, qz^2) + q^2]$. Multiply by a fresh v_i oracle and sum its trace. Equation (7) bounds the z integral by $C(q^2 + qD)$.

For degree zero split the residual into Lemma 5's scalar residual and

$$q^{-1/2}\int_0^\infty k(\omega)\sin(\omega\sqrt{q}z\alpha)(e^{-q\omega^2 r/2}-1)\,d\omega.$$

The latter uses the $\omega^3 k/2$ proposal. The sine and exponential residual moments contribute $q\omega^2 z^2$ and $q^2\omega^4$; the outer q^{-1} and proposal square cancel the displayed powers. Its moment is Cq^2z^2. Finally $\mathbb{E}_\rho Z^2 \le C(1+D)$ by Gaussian maximum entropy, and (6) handles the scalar residual. Since $q \le 2$, these bounds prove (10). □

4 Maintaining the norm without summing all coordinates

A bounded trace alone does not cheaply supply the squared-norm oracle r. We maintain the total squared norm as a separate scalar state.

Lemma 7 (Norm increment)**.** *With the previous scalar oracles, the function*

$$\Delta_z = \frac{\mathbb{E}_G \tanh^2(\sqrt{q}[z\alpha+\sqrt{r}G])}{q} - (z^2\alpha^2+r)$$

has a factory of common expected size $O(T)$ and moment at most

$$Cb(z)^2, \qquad b(z) = q + \min(qz^4, z^2).$$

Consequently $\mathbb{E}_\rho\Delta_Z$ has an oracle of size $O(T)$ and second moment $C(q+D)^2$.

Proof. For the last assertion sample z with density $b(z)/\mathbb{E}_\rho b$ and divide by that density. Equation (7) gives $\mathbb{E}b \le C(q+D)$.

For the local assertion first suppose $qz^2 \le 1$. Two correlated Taylor remainders are useful. In the notation of Lemma 4,

$$\Re Z_s - 1 + (s/\lambda)^2 \sum_{i<j} X_iX_j$$

has mean $\cos(sa) - 1 + (sa)^2/2$ and second moment Cs^8V^4. For $|s|\sqrt{V} \le 1$, expand

$$\mathbb{E}\left[\prod(1+tX_i/\lambda)\prod(1+uX_i/\lambda)\right] = \exp\left(a(t+u) + (\mathbb{E}X^2/\lambda)tu\right).$$

The remaining real terms have degree at least four in each factor. For larger s, the product moment is bounded and the subtracted quadratic has moment $O(1+s^4V^2)$, which is at most $O(s^8V^4)$. The pair sum has an $O(K)$-size expression through $2\sum_{i<j}X_iX_j = (\sum_i X_i)^2 - \sum_i X_i^2$. Its algebraic cancellations use the same realized children; no fourth moment is needed.

Similarly $E_c - 1 + (c/\lambda)\sum_i R_i$ has mean $e^{-cr} - 1 + cr$ and second moment Cc^4. Its small-c series begins at degree two; its large-c bound follows from $\mathbb{E}E_c^2 \le e$ and the second moment of the linear sum.

Use the Fourier identity for $\tanh^2$ and the decomposition

$$\begin{aligned}1 - \cos u\, e^{-c} - (u^2/2 + c) = (1 - \cos u - u^2/2)e^{-c} \\ + (1 - e^{-c} - c) + (u^2/2)(e^{-c} - 1).\end{aligned}$$

Here $u = \omega\sqrt{q}z\alpha$ and the mean $c = q\omega^2 r/2$ is supplied by the r oracle. With proposal $\omega^3 k/2$, the importance coefficient is $2/(q\omega^2)$. The three terms have integrated moments $O(q^2z^8)$, $O(q^2)$, and $O(q^2z^4)$, respectively. Their work is $O(T)$ because $qz^2 \le 1$.

When $qz^2 > 1$, use the direct identity with proposal ωk. Its moment after division by q is at most $C(1+qz^2)/q^2 \le Cz^4$. Subtract independent unbiased products for $z^2\alpha^2$ and the r oracle. This proves the stated cap. □

5 Arbitrary directions and persistent coordinate copies

Let a now be any unit direction in the current Hilbert space, $P = I - aa^*$, and $s = \|h\|^2$. The local helpers are

$$\alpha = \langle a, h\rangle, \qquad v = Ph, \qquad r = s - \alpha^2.$$

A scalar linear functional is sampled with density $|a_i|^2$ over coordinates. If the old coordinate trace is V, its second moment is at most V. The coordinates $v_i = h_i - a_i\alpha$ have trace at most $4V$ and size $O(T)$; r has bounded moment using two independent α copies. These helpers are used only inside the charged residual branches.

Set $\sigma^2 = \mathbb{E}_\rho Z^2$. Gaussian maximum entropy gives

$$\sigma^2 - 1 - \log\sigma^2 \le 2D, \qquad |\sigma - 1| \le \sqrt{2D}, \qquad |\sigma^2 - 1| \le C(\sqrt{D} + D). \tag{11}$$

Define an isometry J by sending h to $(z/\sigma)\langle a, h\rangle$ in degree zero and to Ph in degree one, constant in z. The true linear part is $J[I + (\sigma - 1)aa^*]h$.

For the small residual of Lemma 6, put

$$b_0 = \mathbb{E}_\rho(zR_0/\sigma), \qquad b_i = \mathbb{E}_\rho R_{1,i}.$$

Then $J^*R = ab_0 + Pb$. Sample b_0 with density z^2/σ^2 and b_i with ρ. The resulting trace moments are bounded by the original residual trace. Compute $Pb = b - a\langle a, b\rangle$ with one more sampled scalar functional. Reapplying J gives exactly $(z/\sigma)b_0$ in degree zero and Pb in degree one; no noisy estimate of the identically zero quantity a^*Pb is made.

The isometric embedding

$$F(h) \longmapsto \big(J^*F(h), (I - JJ^*)F(h)\big) = \big([I + (\sigma - 1)aa^*]h + J^*R, (I - JJ^*)R\big) \tag{12}$$

preserves the complete query kernel. Its large part is a literal copy of the old coordinates. Only the residual is projected, at a universal trace/work factor. No fixed anchor or fixed tilt direction is required.

The stored total norm updates through

$$s_{\text{new}} = s + \mathbb{E}_\rho\Delta_Z + (\sigma^2 - 1)\alpha^2.$$

Lemma 7 and (11) give a small-increment oracle with moment $C(q + \sqrt{D} + D)^2$. Thus a full coordinate sum is never computed.

6 Composition and normalization

Write $b = q + \sqrt{D} + D$. For small b, activate the residual with probability proportional to b, and degree $\ell \ge 2$ with probability proportional to b/ℓ^2. Divide an activated residual by its probability. Lemma 6 makes both the trace increment and common expected-size increment $O(b)$. The squared-norm scalar oracle has the same property. Cross terms depend on the true means and are bounded by the corresponding residual norms. The rank-one linear perturbation in (12) is handled the same way.

Whenever a deterministic moment bound exceeds a fixed threshold, average two independent copies. For a vector mean of norm at most one,

$$\int \mathbb{E}\left|(X_i + X_i')/2\right|^2 di \le \tfrac{1}{2}\left(\int \mathbb{E}|X_i|^2 di + 1\right).$$

The scalar norm mean is also bounded. Consequently only $O(1 + \sum b_j)$ resets are needed. Large steps create the residual outright and use polynomially many averages; their logarithmic cost is $O(b)$.

Physical normalization uses $h_{j-1} = u_{j-1}/\sqrt{q_j}$. A local factory returns $u_j/\sqrt{q_j}$, which must be multiplied by $\sqrt{q_j/q_{j+1}}$. The following standard entropy estimate charges this scalar gain.

Lemma 8 (Norm retention)**.** *For a Gaussian entropy tilt of cost D and $\|u\|^2 \le q \le 2$,*

$$\mathbb{E}_\rho \tanh^2(G(u)) \ge e^{-C(q+\sqrt{D}+D)} \|u\|^2 .$$

Proof. Normalize the Gaussian linear form to $Z \sim N(0,1)$ and write $s = \|u\|^2 \le q$. For small $q + \sqrt{D}$, compare $\tanh(\sqrt{s}Z)/\sqrt{s}$ with Z. The residual square is bounded by $C \min(s^2 Z^6, Z^2)$; Lemma 5's entropy argument bounds its tilted mean by $C(q^2 + D)$. The Gaussian quadratic exponential moment also gives $\mathbb{E}_\rho Z^2 \ge 1 - C\sqrt{D}$ in this range. The L^2 triangle inequality therefore gives a relative lower bound $1 - C(q + \sqrt{D})$. For the remaining range, $\tanh^2 t \ge c \min(t^2, 1)$ and $s \le 2$ give the uniform Gaussian small-ball estimate

$$\gamma\{\tanh^2(\sqrt{s}Z)/s \le a\} \le Ca^c$$

for fixed universal constants and $a \in (0,1)$. If its tilted mean is μ, Markov gives tilted mass at least one half to the event with threshold 2μ. Binary entropy data processing yields $D \ge c \log(1/\mu) - C$. Since $q + \sqrt{D}$ is bounded below in this range, the additive constant is absorbed by $C(q + \sqrt{D} + D)$. □

Apply the lemma to a query maximizing q_j. It gives

$$\log^+ \sqrt{q_j/q_{j+1}} \le C(q_j + \sqrt{D_j} + D_j).$$

Vector and squared-norm scalar rescaling therefore have the same total budget. Apply the averaging resets to the normalized state after this physical rescaling, when its vector mean norm and scalar squared-norm mean are at most one. The expected scalar size is initial resources times e^{CB}, with bounded vector trace and scalar norm moments. A sampled final linear functional has second moment at most a constant times the squared readout norm. Average sufficiently many copies and use $O(d+1)$ independent groups. Chebyshev, a binomial tail bound, a union bound on the cube, and a size Markov bound give a finite seed whose majority is correct on every input, with a fixed margin in a fixed majority of the groups.

7 Finite precision without lower bounds on coefficients

The selected formulas may initially contain tiny sampling probabilities, frequencies, or matrix entries. Rounding their original expressions directly is not justified. The following normalization argument supplies finite bits.

For a stored coordinate oracle X_i, let a_i^2 be its average second moment over the Boolean cube and its randomness. Add independent Rademacher noise of variance δa_i^2, with δ inverse polynomial in the number of stored layers. This preserves means and adds only a constant total moment budget. It gives a pointwise independent-noise floor compared with the coordinate's cube RMS. Normalize each internal oracle by its own RMS; zero-RMS coordinates are identically zero on the finite cube and are discarded.

Lemma 9 (Correlated multilinear primitives)**.** *Suppose a primitive is a multilinear polynomial in K distinct child samples, each having an independent Rademacher noise floor at least γ times its cube-averaged second moment. After RMS normalization, its coefficients admit bounds $\exp[O(K \log(K/\gamma))]$ times the normalized parent RMS. The Poisson products and all same-child Taylor remainders above have $O(K)$-size fixed-dimensional vector-algebra representations with internally bounded normalized coefficients. A complete finite tree of these primitives converts once to polynomial-size ordinary real arithmetic formulas.*

Proof. Condition on the input and on base child values bounded in normalized magnitude by $\sqrt{2K}$; the joint cube/randomness probability of this event is at least one half. Walsh

orthogonality in the independent noise signs bounds the translated polynomial's coefficients by its conditional squared norm. Inverting the triangular translation of a multilinear polynomial bounds an original normalized coefficient by $\exp[O(K\log(K/\gamma))]$ times that norm. Integration proves the RMS statement.

A product can be evaluated by keeping fixed degree categories. For the imaginary product minus its linear term use categories 0, 1, even degrees at least two, and odd degrees at least three. For the fourth-order cosine remainder use the analogous six categories separating degrees $0, 1, 2, 3$ from higher even and odd degrees. The exponential remainder uses three categories. Normalize each category by its maximum monomial coefficient. A parent monomial has a unique decomposition between its two disjoint child blocks, so all normalized bilinear transition coefficients have magnitude at most one. This avoids expanding exponentially many monomials.

Any one-hole context in a fixed-dimensional vector formula is affine linear in that hole. A balanced cut therefore expresses the context through a fixed number of affine coefficients and recursively balances the tree with polynomial size and logarithmic depth. Expand the fixed-dimensional bilinear gates into real scalar arithmetic only afterward. This is one global conversion, not a separate expansion at each recursive layer. □

Condition on a primitive's external random choices and let A_θ be its conditional cube RMS, with $A^2 = \mathbb{E}_\theta A_\theta^2$. The ratio satisfies $\mathbb{E}(A_\theta/A)^2 = 1$. An expected-call-count bound excludes large selected ratios with positive probability, simultaneously with the classification and size events. Singular external factors cancel in this RMS normalization. Thus no lower frequency or coefficient cutoff is needed. On a polynomial-size good seed, every normalized coefficient has polynomial logarithmic magnitude, and polynomial-bit rounding preserves the uniformly margined groups.

All arguments extend from sums to standard-Borel direct integrals by Tonelli. Fresh noise is generated only for actually requested coordinates; an uncountable independent random field is unnecessary. A fixed seed requests only finitely many constants. The noise bookkeeping adds a polynomial H factor. Finally the standard arithmetic-formula evaluation and constant-depth composition theorems place the polynomial-size majority in nonuniform PNC^1; $\mathrm{PNC}^1 \subseteq L/\mathrm{poly}$ [1].

8 Fixed-rank blocks

Let $A : \mathbb{R}^r \to \mathcal{H}$ be an isometry with $r \le r_0$, $P = I - AA^*$, and $\alpha = A^*h$. Local helpers for the r coordinates of α, for Ph, and for $s - \|\alpha\|^2$ have trace/work overhead depending only on r_0. For fixed $z \neq 0$, the scalar

$$\alpha_z = \langle Az/\|z\|, h\rangle$$

has a uniform bounded-moment oracle. Substitute $\|z\|\,\alpha_z$ for $z\alpha$ in the preceding factories. Their caps become

$$\min(q^2\|z\|^6, \|z\|^2), \quad \min(q^2\|z\|^4, q\|z\|^2), \quad q + \min(q\|z\|^4, \|z\|^2).$$

The Gaussian entropy estimates hold with constants depending on r_0.

Set $M = \mathbb{E}_\rho ZZ^*$ and $\xi = M^{-1/2}z$. Maximum entropy gives

$$\operatorname{tr} M - \log\det M - r \le 2D, \qquad \left\|\sqrt{M} - I\right\|_{\mathrm{HS}} \le \sqrt{2D}.$$

The isometry J has degree-zero output $\xi^*\alpha$ and degree-one output Ph. Its true linear part is $J[I + A(\sqrt{M} - I)A^*]h$, a charged HS perturbation of the identity. Estimate the ℓth block coordinate of J^*R with density ξ_ℓ^2 and apply the rank-r projector P to the degree-one part. The trace/work overhead is a rank-dependent constant, and only the residual is projected. The scalar norm update adds the quadratic correction $\alpha^*(M - I)\alpha$, using independent copies. Small and large steps, physical rescaling, and Section 7 now prove Theorem 1.

9 Arbitrary Gaussian controls use the same interface

Proposition 10 (An uncentered covariance factorization)**.** *A Gaussian row law $N(m, \Sigma)$ of relative entropy D can be realized by a positive linear change of the input state and an independent one-dimensional Gaussian tilt. Their costs add exactly to D.*

Proof. Set $M = \Sigma + mm^*$, $A = \sqrt{M}$, and $b = M^{-1/2}m$. Then

$$A^{-1}g \sim N(b, I - bb^*), \qquad \|b\| < 1.$$

This law is standard Gaussian perpendicular to b and has mean $\|b\|$ and variance $1 - \|b\|^2$ along its direction. Its entropy is

$$D_1 = -\tfrac{1}{2}\log(1 - \|b\|^2).$$

The covariance-factor cost is $D_0 = \frac{1}{2}(\operatorname{tr} M - \log\det M - n)$. The determinant lemma gives

$$\det M = \det\Sigma(1 + m^*\Sigma^{-1}m), \quad 1 - \|b\|^2 = (1 + m^*\Sigma^{-1}m)^{-1}, \quad D_0 + D_1 = D.$$

Moreover $\|A - I\|_{\mathrm{HS}} \le \sqrt{2D_0}$.

Apply A to the query state before using the one-dimensional factory. For $K = A - I$, the coordinate update has a small-HS oracle: sample a row entry with probability proportional to its square. Its total second moment is at most $\|K\|_{\mathrm{HS}}^2$ times the old trace. The scalar squared-norm increment is

$$h^*(A^*A - I)h, \qquad \|A^*A - I\|_{\mathrm{HS}} \le 2\|K\|_{\mathrm{HS}} + \|K\|_{\mathrm{HS}}^2 .$$

Sample a pair of indices with probability proportional to the squared matrix entry and use independent coordinate calls. Its second moment is bounded by the squared displayed HS norm times the squared old trace. These updates use the same rare-branch and averaging invariants as before.

If $s = \max(1, \|A\|_{\mathrm{op}})$, use normalized state Ah/s and physical scale $q' = qs^2$. Since $\|M\|_{\mathrm{op}} \le C(1 + D)$, $q' \le Cq(1 + D)$. When D is small this charges only $O(q + \sqrt{D} + D)$. When D is large, q' is polynomial in $1 + D$. The one-dimensional Gaussian has mean norm below one and variance at most one, so all its fixed moments are uniformly bounded. The local factories above remain polynomial in q' and D for $q' > 2$: their frequency factors are fixed polynomials, the relative exponential cancellation is unchanged, and the capped bounds can be replaced by the second-moment bound on the tilted scalar. Averaging therefore costs only a polynomial before the physical normalization. Its logarithm is $O(1 + D)$, and Lemma 8 applied to the original Gaussian law charges the final scale change to $O(q + \sqrt{D} + D)$. No switch of state representation is needed. □

10 Bounded quadratic tanh tilts: a positive random-rank mixture

Let $X_i = G(w_i)$ with $\|w_i\|^2 \le q \le 2$, and let

$$V(G) = \tfrac{1}{2}\sum_{i,j} T_{ij}\tanh X_i \tanh X_j, \qquad b = \sum_{i,j}|T_{ij}| \le b_0.$$

If $b = 0$ the law is Gaussian. Otherwise sample $K \sim \mathrm{Pois}(b/2)$ and K ordered pairs with probabilities $|T_{ij}|/b$. Set

$$w_\omega(G) = \prod_{\ell=1}^{K}[1 + \operatorname{sign}(T_{i_\ell j_\ell})\tanh X_{i_\ell}\tanh X_{j_\ell}], \quad Z_\omega = \mathbb{E}_\gamma w_\omega.$$

Every factor is strictly positive. The Poisson product identity gives $\mathbb{E}_\omega w_\omega = e^V$. Thus

$$\rho_T = \int \rho_\omega \, d\pi(\omega), \qquad \rho_\omega = w_\omega \gamma / Z_\omega, \qquad d\pi = Z_\omega \, dP(\omega)/Z_T. \tag{13}$$

Each component changes a subspace of rank at most $2K$ and retains an independent Gaussian complement.

Lemma 11 (Weighted component entropy)**.** *For every fixed $A > 0$ and integer $m \geq 0$, with $D_\omega = \mathrm{KL}(\rho_\omega \| \gamma)$,*

$$\begin{aligned} \mathbb{E}_\pi A^K (K+1)^m &\leq C_{b_0,A,m}, \\ \mathbb{E}_\pi A^K (K+1)^m D_\omega &\leq C_{b_0,A,m} q^2, \\ \mathbb{E}_\pi A^K (K+1)^m \sqrt{D_\omega} &\leq C_{b_0,A,m} q. \end{aligned}$$

Also $D_\omega \leq K(\log 2 + q)$.

Proof. The identity

$$1 + \sigma \tanh x \tanh y = \frac{\cosh(x + \sigma y)}{\cosh x \cosh y}$$

and $\log \cosh x \leq x^2/2$ give $Z_\omega \geq e^{-Kq}$ by Jensen. The product is at most 2^K, and $Z_T \geq e^{-b/2}$. Hence all fixed exponential rank moments are finite uniformly in q.

For one factor h, $\mathbb{E}_\gamma \|\nabla h\|^2 \leq 4q^2$. The product rule, the bound $h \leq 2$, and Gaussian Poincaré give

$$\mathrm{Var}_\gamma(w_\omega) \leq K^2 4^K q^2.$$

The KL-to-chi-square inequality therefore yields

$$Z_\omega D_\omega \leq \frac{\mathrm{Var}(w_\omega)}{Z_\omega} \leq K^2 (4e^q)^K q^2.$$

Integrate under the Poisson law and use Cauchy–Schwarz for the square-root estimate. Finally $\log(w_\omega / Z_\omega) \leq K(\log 2 + q)$ pointwise. □

The local block factories have *polynomial* rank dependence before any large-step reset. More precisely, for a universal fixed integer p, balancing coordinate work gives residual traces bounded by

$$C(1+r)^p (q^2 + D),$$

and scalar norm-increment moments bounded by $C(1+r)^p(q^2 + D + D^2)$. To verify this quantitative point, block projection helpers have polynomial rank moments and work. Tensor averaging cancels the true-norm exponential before the degree sum, leaving only fixed powers of those moment bounds. All Fourier moment orders are fixed. Taking entropy dual parameter $1/(8r)$ gives

$$\mathbb{E}_\rho \min(q^2 \|Z\|^6, \|Z\|^2) \leq C(rD + q^2 r^3), \quad \mathbb{E}_\rho \min(q \|Z\|^4, \|Z\|^2) \leq C(rD + qr^2).$$

Whitened projection coordinates have normalized squared densities, so no inverse eigenvalue enters a moment estimate. There are only finitely many operations of fixed polynomial degree. Increasing p absorbs common-work balancing and projection costs.

It is essential to integrate these *unreset* factories before paying component-dependent normalization costs. Lemma 11, including $D_\omega^2 \leq C(K+1)D_\omega$, gives integrated residual trace and norm-increment moment $O_{b_0}(q^2)$.

Here is the common isometry explicitly. Let J_ω be the normalized block isometry for component ω, and let $S_\omega - I$ be its HS linear correction. On the direct integral with measure π, put

$$Jh = (J_\omega h)_\omega, \qquad \widetilde{R}_\omega = R_\omega + J_\omega (S_\omega - I) h.$$

Then $J^*J = I$ and $F = Jh + \widetilde{R}$. The additional linear residual has integrated HS squared norm at most $2\mathbb{E}_\pi D_\omega = O_{b_0}(q^2)$. Apply the exact embedding

$$F \longmapsto (h + J^*\widetilde{R}, (I - JJ^*)\widetilde{R}).$$

The component adjoints, mixture integration, and reapplication of J_ω are sampled and balanced to common coordinate work; each adds only a fixed rank polynomial, absorbed by Lemma 11. Thus the old state is again a literal copy with a residual of second moment $O_{b_0}(q^2)$. The total-norm oracle uses the averaged component increment and has the same bound.

Rare creation with probability proportional to q now gives $1 + O_{b_0}(q)$ work and moment increments. Convexity of relative entropy gives $D(\rho_T\|\gamma) \le \mathbb{E}_\pi D_\omega = O_{b_0}(q^2)$, so physical scaling has the same cost. The global precision argument remains unchanged. This proves the quadratic-tilt extension stated after Theorem 1.

At a minimum hard-margin final readout, its dual coefficients β_x satisfy $\|\beta\|_1 \le \|v\|^2/a$ for margin a, so the terminal quadratic costate $T = \beta\beta^*$ has bounded entrywise norm whenever the action is bounded. Earlier variational costates need not have a known uniform bound. The extension therefore handles that terminal family but does not certify the entire optimal posterior path.

11 Normalized ridge differences and a Gaussian envelope

We prove Theorem 2. For each normalized ridge or ridge difference, let a_i be the corresponding unit direction $w_i/\|w_i\|$ or $(w_i - z_i)/\|w_i - z_i\|$, and write $A_i = g^*a_i$. The Lipschitz bounds for tanh and its derivative give

$$|\psi_i(g)| \le |A_i|, \qquad \|\nabla\psi_i(g)\| \le 1 + C|A_i|. \tag{14}$$

For a difference, its gradient equals

$$a_i \operatorname{sech}^2(g^*w_i) + z_i \frac{\operatorname{sech}^2(g^*w_i) - \operatorname{sech}^2(g^*z_i)}{\|w_i - z_i\|}.$$

Thus the constant is uniform through arbitrarily small denominators, since $q \le 2$.

Put $V = \frac{1}{2}\sum S_{ik}\psi_i\psi_k$. The case $t = 0$ is Gaussian. For $t > 0$, define

$$P = \frac{1}{2t}\sum_{i,k} |S_{ik}|(a_i a_i^* + a_k a_k^*), \qquad Q(g) = \tfrac{t}{2} g^* P g.$$

Then $P \succeq 0$, $\operatorname{tr} P = 1$, and $|V| \le Q$. Change the Gaussian reference to

$$d\gamma_Q \;\propto\; e^{-Q}\, d\gamma, \qquad C_Q = (I + tP)^{-1}.$$

Its covariance square root differs from the identity in HS norm by $O(t)$, and its Gaussian entropy is $O(t^2)$.

Choose a pair $I = (i, k)$ with probability $|S_{ik}|/t$, and set

$$h_I = \tfrac{1}{4}(A_i^2 + A_k^2 + 2\operatorname{sign}(S_{ik})\psi_i\psi_k), \qquad H = \mathbb{E}_I h_I.$$

Equation (14) gives

$$0 \le h_I \le (A_i^2 + A_k^2)/2, \qquad V + Q = tH, \qquad H \le g^*Pg. \tag{15}$$

Draw $K \sim \mathrm{Pois}(1)$ and independent pairs $I_1, \dots, I_K$. With

$$w_\omega = \prod_{\ell=1}^{K}(1 + th_{I_\ell}), \quad Z_\omega = \mathbb{E}_{\gamma_Q} w_\omega, \quad \rho_\omega = w_\omega \gamma_Q / Z_\omega,$$

all factors and Z_ω are at least one. The exact identity $\mathbb{E}_\omega w_\omega = e^{tH}$ gives

$$\rho = \int \rho_\omega \, d\pi(\omega), \qquad d\pi = Z_\omega \, dP(\omega)/\overline{Z}, \qquad \overline{Z} = \mathbb{E}_{\gamma_Q} e^{tH} \geq 1. \tag{16}$$

In whitened γ_Q coordinates each component changes a subspace of dimension at most $4K$ and has an independent standard Gaussian complement.

Lemma 12 (Weighted component moments)**.** *Fix $A > 1$ and an integer $m \geq 0$. Let z_ω be the changed block in whitened γ_Q coordinates, of rank $r_\omega \leq 4K$. For sufficiently small t_0,*

$$\mathbb{E}_\pi A^K \mathbb{E}_{\rho_\omega} \|z_\omega\|^m \leq C_{A,m}.$$

The same holds with any fixed polynomial in K inserted.

Proof. Cancel Z_ω against the mixture weight and use $w_\omega \leq \prod_{\ell=1}^K [1 + t(A_{i_\ell}^2 + A_{k_\ell}^2)/2]$. All these linear forms have Gaussian variance at most one. For any ℓ selected forms L_a, Gaussian Hölder and Cauchy–Schwarz give

$$\mathbb{E} \|z_\omega\|^m \prod_{a=1}^{\ell} L_a^2 \leq C_m (K+1)^{m/2} \sqrt{(4\ell - 1)!!} \leq C_m (K+1)^{m/2} C^\ell (2\ell - 1)!!.$$

This holds despite the dependence of the block subspace on ω. Expanding the product and summing $K \sim \mathrm{Pois}(1)$, without its polynomial factor for the moment, gives

$$\mathbb{E}_K A^K \sum_{\ell \leq K} \binom{K}{\ell} (Ct)^\ell (2\ell - 1)!! = e^{A-1} \mathbb{E}_{Z \sim N(0,1)} e^{CAtZ^2}.$$

A larger fixed exponential weight absorbs every fixed polynomial in K. The final Gaussian integral is finite for sufficiently small t_0; division by $\overline{Z} \geq 1$ proves the claim. □

We now improve the direct component-entropy estimate by using one polar isometry for the whole mixture. Let h be the normalized incoming state, with $\|h(x)\| \leq 1$. In the component Fock representation write

$$F_\omega(h) = \tanh(\sqrt{q}\, g^* h)/\sqrt{q}, \qquad L_\omega h = g^* h, \qquad R_\omega = F_\omega - L_\omega.$$

The block factories of Section 8, before their entropy caps are applied, have vector residual traces and scalar squared-norm increment moments bounded by

$$C(1+K)^p q^2 (1 + \|z_\omega\|^m) \tag{17}$$

for fixed integers p, m. Degree weights include the scalar work and later projections. Indeed the scalar residual uses $q^2 \|z\|^6$, the degree-one residual uses fixed fourth moments, and the norm increment uses fixed eighth moments. The damped tensor sum introduces no degree-dependent moment order in z. Multiplying by a block coordinate adds only two to the fixed moment order. Lemma 12 therefore gives integrated residual traces and norm-increment moments $O(q^2)$ at common work $O(T)$.

In these factories the helpers $C_Q^{1/2} h$ cost only a constant factor: $C_Q^{1/2} - I$ has HS norm $O(t)$. They are used inside correction factories; the old-state copy is not first updated by $C_Q^{1/2}$. Thus this operation incurs no repeated $O(t)$ charge.

On the direct-integral feature space set

$$Lh = (L_\omega h)_\omega, \qquad \Sigma = L^* L = \mathbb{E}_\rho g g^*, \qquad J = L\Sigma^{-1/2}.$$

The two Gaussian quadratic comparisons from $|V| \le Q$ give $cI \preceq \Sigma \preceq CI$ uniformly. Gaussian maximum entropy at fixed uncentered second moment gives

$$\tfrac{1}{2}\operatorname{tr}(\Sigma - I - \log \Sigma) \le D, \qquad \left\|\Sigma^{1/2} - I\right\|_{\mathrm{HS}}^2 \le 2D.$$

The latter follows from $(\sqrt{s}-1)^2 \le s - 1 - \log s$. Consequently $J^*J = I$ and $\Sigma^{-1/2} - I$ has HS norm $O(\sqrt{D})$. These assertions extend from finite-dimensional row spans by the same Gaussian finite-dimensional approximation as the earlier interfaces. The exact isometric embedding gives

$$F = Lh + R \longmapsto (\Sigma^{1/2}h + J^*R,\ (I - JJ^*)R). \tag{18}$$

Both projections in (18) have economical scalar realizations. Write U_ω for the block isometry in whitened coordinates and $P_\omega = I - U_\omega U_\omega^*$. The adjoint involves only Fock degrees zero and one:

$$L_\omega^* R_\omega = C_Q^{1/2}\left[U_\omega \mathbb{E}_{\rho_\omega, z}(z R_{\omega,0}(z)) + P_\omega \mathbb{E}_{\rho_\omega, z} R_{\omega,1}(z)\right].$$

Block helpers and coordinate integrations are sampled as in Section 8; their costs have only fixed polynomial rank dependence. The extra factor z is covered by Lemma 12. Averaging components, then applying $\Sigma^{-1/2}$ through its identity-plus-HS representation, supplies J^*R with trace $O(q^2)$ and common work $O(T)$. Applying the same degree-zero and degree-one formulas forward supplies JJ^*R with the same bounds. All these maps are applied only to corrections, so they do not multiply the old-state cost by a constant at each layer.

The linear part $\Sigma^{1/2}h$ uses the existing HS vector and squared-norm calculus, at cost $1 + O(\sqrt{D} + D)$. Its remaining scalar norm correction is exactly

$$\mathbb{E}_\rho\left[\tanh^2(\sqrt{q}\, g^*h)/q - (g^*h)^2\right],$$

whose factory has moment $O(q^2)$ by (17). Rare creation of the nonlinear vector and norm corrections with probability proportional to q therefore gives the additional work/moment increment $O(q)$.

For completeness the actual entropy satisfies $D \le Ct^2$. Gaussian logarithmic Sobolev gives

$$D \le \tfrac{1}{2}\mathbb{E}_\rho \|\nabla V\|^2 \le Ct^2,$$

using (14), the quadratic envelope, and $Z \ge e^{-t/2}$. Physical normalization is charged by Lemma 8 using the actual D, not the component entropies. Its cost is $O(q + \sqrt{D} + D)$. Large stages use the earlier reset. The new scalar nodes are only existing projection, HS, mixture, and residual nodes with the required trace/work bounds, so the moment-floor finite-bit argument is unchanged. Identically zero corrections are omitted before division. This proves Theorem 2 with actual entropy in the accumulated budget.

For example, the potential

$$\frac{t}{2}\frac{[\tanh(g^*w) - \tanh(g^*z)]^2}{\|w - z\|^2}$$

is covered uniformly as w approaches z, although its expanded raw coefficients diverge. A general covariance-normalized feature direction can involve a complicated combination of many ridges. We have not proved that every such direction admits a normalized-ridge representation with controlled coefficient mass.

12 Arbitrary first and last row laws

The initial feature space has dimension $d + 1$, so an arbitrary first row law can be handled directly. The final layer can instead transfer a truncated density into the linear readout. Neither argument applies to a general internal layer.

Lemma 13 (Initialization under an arbitrary first row law)**.** *For $\rho_1 \ll \gamma_{d+1}$ with entropy $D_1 < \infty$, set*

$$h_1(x)(g) = \tanh(g^* u_0(x)), \qquad r(x) = \|h_1(x)\|^2_{L^2(\rho_1)}, \qquad q_* = \max_x r(x).$$

There are scalar coordinate oracles for $h_1/\sqrt{q_*}$ and a separate oracle for r/q_*, with bounded integrated trace and scalar second moments and common expected work $\mathrm{poly}(d+D_1+2)\exp(CD_1)$. *The finite-bit construction used above continues to apply.*

Proof. Entropy duality at parameter $1/4$ gives

$$\mathbb{E}_{\rho_1} \|g\|^2 \le 4D_1 + 2(d+1)\log 2.$$

Every reference projection $g^* u_0(x)$ has variance $(d+1)/d \in [1,2]$. For small $r = r(x)$, Markov's inequality gives ρ_1-probability at least $3/4$ to $|\tanh(g^* u_0(x))| \le 2\sqrt{r}$. Its Gaussian probability is at most $C\sqrt{r}$. Binary relative entropy therefore implies

$$D_1 \ge \tfrac{3}{4}\log(1/(C\sqrt{r})) - \log 2.$$

The remaining values of r are harmless. Uniformly over the cube,

$$r(x) \ge ce^{-CD_1}, \qquad ce^{-CD_1} \le q_* \le 1. \tag{19}$$

For fixed g, the linear form $g^* u_0(x)$ has size $O(d+1)$ and squared magnitude at most $2\|g\|^2$. The Fourier–Poisson factory, balanced conditionally at each frequency, produces a tanh oracle with common expected size $O(d+1)$ and second moment at most $C\|g\|^2$. Indeed sparsifying a conditional factory by its work factor $1+\omega^2 V_g$, and rescaling its active output, makes its work constant; the optimal frequency proposal and the finite weighted first Fourier moment then give this work–moment bound. The case $g = 0$ is the zero oracle.

Integrating over ρ_1 gives trace second moment $O(d+D_1+1)$. Averaging makes it bounded at polynomial work. For the norm oracle we do *not* square two tanh factories at the same g, which would unjustifiably require a fourth moment. Instead use a direct Fourier-increment factory for $\tanh^2 z = 1 - \mathrm{sech}^2 z$. Its exponentially decaying Fourier transform has finite weighted first moment, so the same work–moment estimate holds. Sampling g and averaging therefore gives a separate oracle for $r(x)$ using only the displayed second-moment bound.

Divide the coordinate oracles by $\sqrt{q_*}$ and the norm oracle by q_*. By (19), an averaging reset restores the bounded-moment invariants at the claimed cost. The means are exactly $h_1/\sqrt{q_*}$ and its squared norm r/q_*, so this is the common-scale interface of the preceding compiler.

There is no coefficient-size premise. Before the first scalar transform, cube-RMS normalization changes the deterministic linear-form coefficients to $g_i/\|g\|$, all of magnitude at most one. The subsequent Fourier, Poisson, importance, and averaging nodes use the same moment-floor and conditional-RMS normalization argument as before. Its integrated trace and common-work hypotheses have just been verified. A good finite realization contains finitely many constants and admits the same polynomial-bit rounding. No efficient sampling or density-evaluation claim is made. □

Lemma 14 (Deleting an arbitrary final density)**.** *Let $\rho = r\gamma$ have entropy $D < \infty$, let $|F_x| \le 1$, and suppose $v \in L^2(\rho)$ has norm V and signed margin $\eta > 0$ at every query. If $D > 0$, there is a readout under γ with margin at least $\eta/2$ and norm at most*

$$V \exp\left[12\left(\sqrt{D}\,V/\eta + DV^2/\eta^2\right)\right].$$

For $D = 0$ the original readout and margin are unchanged.

Proof. For $M > 1$, the nonnegative integrand $r \log r - r + 1$ gives

$$\rho\{r > M\} \le \frac{D}{\log M - 1 + 1/M}.$$

Put $w = vr\,\mathbf{1}_{\{r \le M\}}$. Then

$$\|w\|^2_{L^2(\gamma)} \le MV^2, \qquad |\mathbb{E}_\rho v F_x - \mathbb{E}_\gamma w F_x| \le V\sqrt{\rho\{r > M\}}.$$

Take $u = DV^2/\eta^2$ and $s = \log M = 24(\sqrt{u} + u)$. The elementary bounds $s - 1 + e^{-s} \ge s^2/3$ for $0 \le s \le 1$ and $s - 1 + e^{-s} \ge s/4$ for $s \ge 1$ make the tail at most $\eta^2/(4V^2)$. This proves the result. Entropy zero means that the two laws coincide. □

Initialize with Lemma 13, run the proved internal compiler, and replace the final row law by its Gaussian reference using Lemma 14. The final Gaussian layer has incoming squared norm at most one, so adds only a constant to the budget. Apply the usual readout sampling, majority amplification over the cube, and finite-bit conversion. The enlarged norm and halved margin give exactly Corollary 3. At one hidden layer, apply the readout directly to Lemma 13. The density deletion preserves the classifier, not the original action; any near-baseline occupation estimate is applied to the original path.

13 What remains for unrestricted critical paths

The proof represents the non-Gaussian features directly; replacing them by a Gaussian is not sufficient in general. Even three equal-norm queries admit entropy-D examples whose covariance-matched Gaussian kernels have optimal common feature error of order $\sqrt{qD}$. The discrepancy can affect useful polynomial-norm readouts. Entrywise kernel errors of order $q^2\sqrt{D} + qD$ do not repair that problem when weak modes matter.

The remaining issue is a dimension-free treatment of an internal row law with general conditional dependence. Talagrand transport writes the changed Gaussian row as $Z + b(Z)$ with $\mathbb{E}\,\|b\|^2 \le 2D$, giving a small HS *linear* perturbation. It does not by itself give economical scalar formulas for the nonlinear feature change. The entropy chain rule, a covariance match, or a bounded trace cannot be substituted for that missing argument. The present theorem therefore supplies a genuinely non-Gaussian class with controlled formula complexity while leaving the full critical-posterior question open.

References

[1] S. Datta, M. Mahajan, B. V. Raghavendra Rao, M. Thomas, and H. Vollmer. *Counting Classes and the Fine Structure between NC1 and L*. 2010. Primary manuscript: https://www.cse.iitm.ac.in/~bvrr/conf/mfcs10.pdf. We use Proposition 2.4 and the constant-depth composition results in Theorems 4.1–4.2.